

Statistics

- Mean (माध्य)
- Mean deviation (माध्य विचलन)
- Median (मध्याका)
- Mode (बहुलक)
- Range (परास)
- Coefficient of range (परास गुणांक)
- Standard deviation (मानक विचलन)
- Variance (विसरण)
- Coefficient of variation (विसरण गुणांक)

-PYQs



$$\text{Mean } (\bar{x}) = \frac{\sum x}{N}$$

$$\text{M.D } (\bar{x}) = \frac{|x_i - \bar{x}|}{n}$$

2, 1, 7, 10, 5, 15

$$\text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value.}$$

15-1

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ observation} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation}}{2}$$

NCEB

$$\text{Coefficient of Range} = \frac{L-S}{L+S}$$

15-1
15+1 = 16

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$\text{Variance}(\sigma^2) = \frac{\sum (x_i - \mu)^2}{N}$$

$$\text{Coefficient of variation} = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

$$\text{Standard deviation}(\sigma) = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

= $\sqrt{\text{Variance}}$



2, 3, 3, 5, 5, 10, 7, 7, 7, 1

$$1+0+0+2+2+7+4+4+4+2$$

10

Mean (माध्य) $\rightarrow \frac{\text{Sum of all terms}}{\text{no. of terms}} = \frac{50}{10} = 5$

Median (मध्यिका) $\rightarrow$ 1 2 3 3 5 5 7 7 7 10

(5)

Mode (बहुलक) $\rightarrow 7$

Range (परास) $\rightarrow 10-1 = 9$

Coefficient of range (परास गुणांक) $\rightarrow \frac{\text{Max} - \text{Min}}{\text{Max} + \text{Min}} = \frac{10-1}{10+1} = \frac{9}{11}$

Mean deviation about mean (5) $\rightarrow \frac{3+2+2+0+0+5+2+2+2+4}{10} = 2.2$

Mean deviation about median (3) $= 2.2$

2, 3, 3, 5, 5, 10, 7, 7, 7, 1

$$\text{Mean} = \underline{5}$$

$$\frac{20}{50}$$

Variance (विसरण)

$$= \frac{3^2 + 2^2 + 2^2 + 5^2 + 2^2 + 2^2 + 2^2 + 4^2}{10} = \frac{9 + 4 + 4 + 25 + 4 + 4 + 4 + 16}{10} = \frac{70}{10} = 7$$

Standard deviation (मानक विचलन) = $\sqrt{\text{Variance}} = \sqrt{7}$

Coefficient of variation (विसरण गुणांक) = $\frac{\text{Standard deviation}}{\text{mean}} = \frac{\sqrt{7}}{5}$



✓ In a moderately skewed distribution, the median is 40 and the mean is 42.
Using these values, find the approximate value of the mode.

$$\begin{aligned}\text{Mode} &= \frac{2\text{Median} - 3\text{Mean}}{3} \\ &= \frac{120 - 126}{3} \\ &= \underline{\underline{-2}}\end{aligned}$$

Mean

Median

Mode

✓

✓

X

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✓ Find the mean, median and mean deviation about the mean of the following data:

$$6 \times 3 + 12 + 8 + 14 + 16 + 10 + 32$$

42

Size (x):	<u>1</u>	<u>3</u>	<u>5</u>	<u>7</u>	<u>9</u>	<u>11</u>	<u>13</u>	<u>15</u>
Frequency (f):	3	3	4	14	7	4	3	4

→ 4+2

21 22

Mean → 8 → $\bar{x}$

$$\underline{\underline{MD_{\text{man}}}} \Rightarrow \frac{21 + 15 + 12 + 14 + 7 + 12 + 15 + 28}{42 \leftarrow} = \frac{\sum_{i=1}^n f_i |(\bar{x} - x_i)|}{\sum f}$$

Medi = 7

mode = 7

$$\underline{\underline{MD_{\text{median}}}} = \frac{\sum f |x_i - M|}{\sum f} \rightarrow M \rightarrow \text{median.}$$



Median = ? Mode = ?

✓ Calculate Mean, variance, standard deviation of the following data :

Class interval	Frequency
4 - 8	<u>3</u>
8 - 12	<u>6</u>
<u>12 - 16</u> →	<u>4</u>
<u>16 - 20</u> →	<u>7</u>
	<u>20</u>

$$\text{Mean} = \frac{\sum fx}{\sum f} = \frac{18 + 60 + 56 + 126}{20} = \frac{260}{20} = \underline{13}$$

$$\text{variance } (\sigma^2) = \frac{99 \times 3 + 9 \times 6 + 4 + 175}{20} = \frac{300}{20} = \underline{15}$$

$$\text{Standard deviation } (\sigma) = \sqrt{\text{variance}} = \sqrt{15}$$

$$\text{coef. of } \underline{\text{varian}} = \frac{\text{S.D.}}{\text{Mean}} = \frac{\sqrt{15}}{13} \times 100 \%$$

Median → 12 - 16

Mode → 16 - 20

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mathsmaniassc

Median of grouped data with class interval.

$$\rightarrow \text{Median} = l + \left[\frac{\frac{n}{2} - c}{f} \right] \times h$$

Median of grouped data with class interval. Median = $l + \left[\frac{\frac{n}{2} - C}{f} \right] h \rightarrow p - L$

Marks	Number of students	Cf
0 - 20	6	6
20 - 40	20	26
40 - 60	37	63
60 - 80	10	73
80 - 100	7	80

$l = 40$
 $n = 80$
 $f = 37$
 $h = 20$
 $C = 26$

$n = \sum f = 80$
 $40 \quad 41$

40 - 60

- a) 39
- b) 63
- c) 35
- d) 45

✓ Median of grouped data with class interval.

Class Intervals	No. of girls (f_i)
120-130	2
130-140	8
140-150	12
<u>150-160</u>	20
160-170	8

a) 137
 b) 145
 c) 152
 d) 161

a) 137
 b) 152
 c) 157
 d) 161

→ 22 → C

$h = 10$

$$\text{Median} = \underline{l} + \left[\frac{\frac{n}{2} - C}{f} \right] \times h$$

150 - 155

$$n = \sum f = 50 \rightarrow \underline{25^{th} \text{ } 26^{th}}$$

✓ Median of grouped data with class interval.

Weekly Expenditure	No. of families (f_i)
0 - 1000	<u>34</u>
1000 - 2000	<u>12</u>
2000 - 3000	43
→ <u>3000 - 4000</u>	<u>60</u>
4000 - 5000	<u>51</u>

$$C = \underline{89}$$

$$h = \underline{1000}$$

- a) 2500
 b) 3900 α
 ✓ c) 3100
 d) 4100

$$\text{Median} = l + \left(\frac{\frac{n}{2} - C}{f} \right) h$$

$$\underline{\underline{3000 - 3500}}$$



$$n = \underline{\Sigma f} = 200 \quad \underline{100} \quad \underline{101}$$



✓ Mode of grouped data with class interval (CGL 2023 Mains)

$$\checkmark \text{ Mode} = \underline{L} + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \underline{h} \quad \text{or} \quad L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

where,

- L is the lower limit of the modal class
- h is the size of the class interval
- f_1 is the frequency of the modal class
- f_0 is the frequency of the class preceding the modal class
- f_2 is the frequency of the class succeeding the modal class

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✓ Mode of grouped data with class interval.

Height (in cm)	Number of students
120 - 125	3
125 - 130	5 f_0
$\leftarrow$ <u>130 - 135</u> $\longrightarrow$	11 f_1
135 - 140	<u>6</u> f_2
140 - 145	5
Total	30

$h=5$

- a) 127
~~b) 134~~
 c) 136
 d) 137

- a) 127
 b) 131
~~c) 134~~
 d) 136

$$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

$$132.5 - 135^*$$

Mode of grouped data with class interval.

formula $\rightarrow$ exact $\times$
 $\hookrightarrow$ estimate

Class Interval	Frequency
10 - 20	8
20 - 30	15
30 - 40	12
40 - 50	5
Total	40

Handwritten notes on the table:

- A red arrow points from the frequency 8 to the frequency 15.
- A red arrow points from the frequency 15 to the frequency 12.
- The frequency 12 is circled in red.
- Next to the class interval 20 - 30, the text 25-30 is written in red.
- Next to the class interval 20 - 30, a red arrow points from the interval to the frequency 15.



Mode of grouped data with class interval.

Class Interval	Frequency
10 - 25	2
25 - 40	3
<u>40 - 55</u>	<u>7</u>
55 - 70	<u>6</u>
70 - 85	6
85 - 100	6

(47.5 - 55)

$$\text{Mode} = l_f \left[\frac{f_1 - f_0}{2f_1 - f_0 + f_2} \right] h$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$



✓ SSC CGL MAIN 2023

A set of data presented in the form of a frequency distribution table with class intervals and their respective frequencies had a mode of 48.5. The lower boundary of the modal class was 46.5; the frequency of the modal class was 34; the frequency of the class interval just preceding the modal class was 32 and the frequency of the class interval just succeeding the modal class was 25. What was the width of the modal class?

✗ 1. 10

✗ 2. 12

✓ 3. 11

✗ 4. 10.5

$$\text{mode} = 48.5$$

$$l = 46.5$$

$$f_1 \rightarrow 34$$

$$f_0 = 32$$

$$f_2 = 25$$

$$h = ?$$

~~✗~~

$$48.5 = 46.5 + \left(\frac{2}{68 - 32 - 25} \right) h$$

$$2 = \frac{2 \times 11}{11}$$

$$11 = h$$

f_0

f_1

f_2



✓ SSC CGL MAIN 2022

Find the mode of the given distribution (rounded off to two decimal places)?

Class Interval	5-10	10-15	15-20	20-25	<u>25-30</u>	30-35
Frequency	8	7	6	9	<u>11</u>	10

बहुलक ?

25 27.5

27.5 30

A) 40.25

~~B) 28.33~~ →

C) 30.33

D) 35.25

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✓ SSC CGL MAIN 2022

Find the median of the following numbers? मधिका ?

32, 25, 33, 27, 35, 29 & 30

↑ ↑ ↑ ↑ ↑ ↑

A) 27

B) 32

✓ C) 30

D) 29



SSC CGL MAIN 2022

Find the standard deviation of the following data
(rounded off to two decimal places) मानक विचलन ?

5, 3, 4, 7

$$\frac{19}{4} = \underline{4.75}$$

$$\sqrt{\frac{(\underline{0.25})^2 + (\underline{1.75})^2 + (\underline{1.75})^2 + (\underline{2.25})^2}{4}}$$

$$\underline{0.25} - \underline{2.25}$$

~~A) 1.48~~

B) 4.12 α

C) 2.45 $\leftarrow 2$

D) 3.21 $\leftarrow 7$

Mean deviation



SSC CGL MAIN 2022

The mean of a sample data is 60 and the median is 48. Find the mode of this distribution?

ढाँडा?

- A) 36
- B) 18
- ☒ C) 24
- D) 48

Mode = $3 \text{ Median} - 2 \text{ Mean}$
 $= 144 - 120$
24

1, 1, 1, 2 .

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SSC CGL MAIN 2022

✓ What will be the mode of the following data?

13, 15, 31, 12, 27, 13, 27, 30, 27, 28, 16

A) 28

B) 31

C) 30

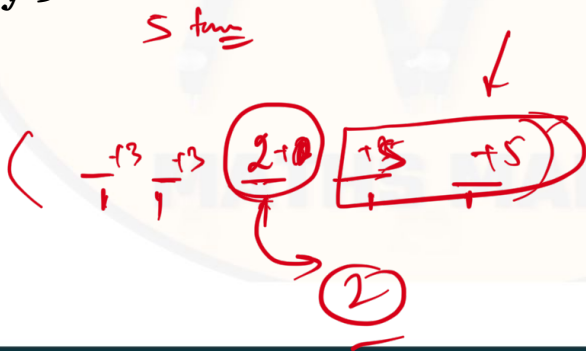
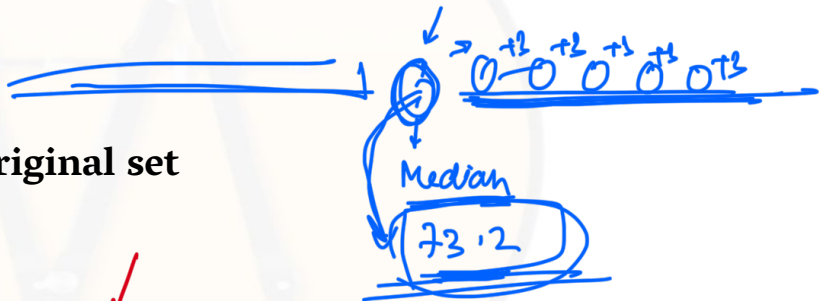
D) 27 → Mode → गुणित

SSC CGL MAIN 2022

The median of a set of 11 distinct observation is 73.2. if each of the largest five observation of the set is increased by 3, then the median of the new set?

11 विशिष्ट अवलोकनों के एक सेट का माध्य 73.2 है। यदि समुच्चय के सबसे बड़े पाँच प्रेक्षणों में से प्रत्येक को 3 से बढ़ा दिया जाए, तो नए समुच्चय की माध्यिका क्या होगी?

- A) Is 3 times that of the original set
- B) Is increased by 3
- ☒ C) Remains the same as that of the original set
- D) Is decreased by 3



SSC CGL MAIN 2022

What is the mode of the given data?

5, 7, 9, 7, 3, 7, 5, 7, 8, 6, 7

- ☒ A) 7
- B) 6
- C) 5
- D) 3

SSC CHSL MAIN 2023

Calculate the mode from the following data.

Days of confinement	6	7	8	9	10
Number of patients	4	6	<u>7</u>	5	3

1. 7

2. 5

3. 6

~~4.~~ 8

SSC CHSL MAIN 2023

H.W.

✓ Find the mode of the given data.

21, 15, 16, 48, 6, 15, 44, 16, 6, 5, 15, 16, 21, 6, 15 and 43

1. 16
2. 21
3. 6
4. 15



SSC CHSL MAIN 2023

The runs scored by 11 players in the cricket match are as follows.

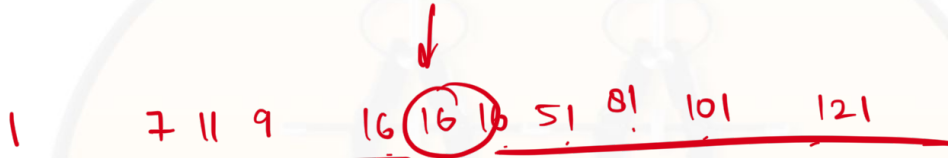
7, 16, 121, 51, 101, 81, 1, 16, 9, 11, 16. Calculate the median of the data.

1. 51

2. 11

3. 22

✓ 4. 16



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SSC CGL MAIN 2023

The marks obtained by 15 students out of a maximum of 25 in a test are given as 13, 11, 16, 15, 18, 12, 13, 14, 10, 22, 15, 21, 20, 17 and 24. Find the product of the modes of this set of data.

एक परीक्षा में अधिकतम 25 में से 15 छात्रों द्वारा प्राप्त अंक 13, 11, 16, 15, 18, 12, 13, 14, 10, 22, 15, 21, 20, 17 और 24 हैं। डेटा के इस सेट के मोड का उत्पाद।

(a) 182

(b) 192

☒ (c) 195

(d) 210

13 13

11

16

15 15

18

12

14

10

22

15 × 13 =

195

21

20

17

24



Find the median of the following data.

15, 30, 20, 10, 25, 35, 18, 21, 28

- (a) 18
- (b) 25
- (c) 30
- (d) 21



SSC CHSL MAIN 2022

H.W.

9th Jan

Find the median of the following data.

129.75, 116.6, 12.3, 114.1, 136.6, 118.7, 13.5, 121.3

(a) 116.60

~~(b) 117.65~~

(c) 127.80

(d) 118.75

$$\begin{array}{r} 116.6 \\ 118.7 \\ \hline 235.3 \\ \hline 2 \end{array}$$

$$\begin{array}{r} 117.65 \\ \hline \end{array}$$

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THANK YOU !

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